

Max Heap

→ Insert 9: → Insert 2: → Insert 1: → Insert 10:

First value becomes Root

Transition 1: Swap 10 & 2 → Transition 2: Swap 10 & 9:

Insert 10 into Next available Position. Bubble it upward by Swapping with 2 and then 9

→ Insert 8: → Insert 3: → Insert 7: → Swap 7 & 3:

Insert 4: → Swap 4 & 2: → Insert 6: → Swap 6 & 4:

→ Insert 5: → Delete Root (10): → Transition 1:

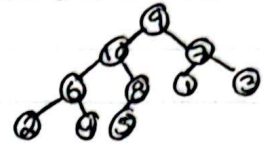
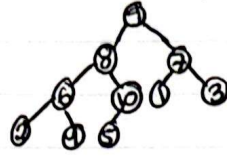
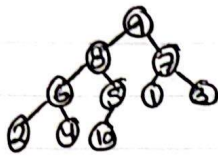
Have last element to Root

→ Transition 2:

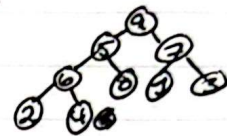
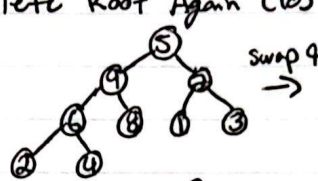
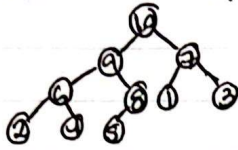
Delete Root 10. Replace it with the last element, 5. Bubble 5 downward by Swapping with the larger Child each time: first 4, then 8

Max Heap

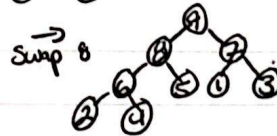
→ Insert 10 Back: → Transition 1: Swap 5 & 10 → Transition 2: Swap 10 & 8



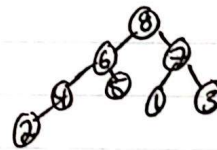
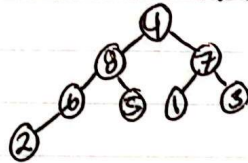
→ Transition 3: Swap 10 & 9 → Delete Root Again (10):



Reinsert 10. Goes Back to Root



→ Delete Root one more time (9): → Transition 1: Swap 4 & 6



Final Max Heap

